Image Formation

Discussion #2

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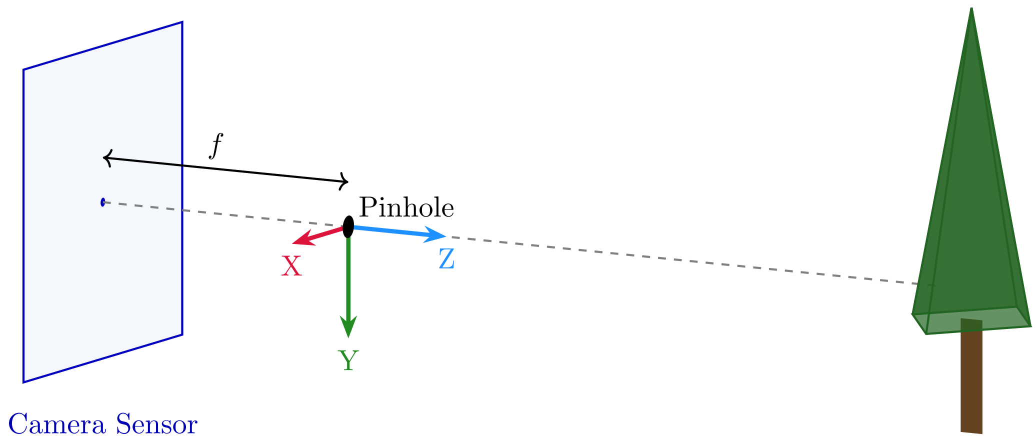
Topics

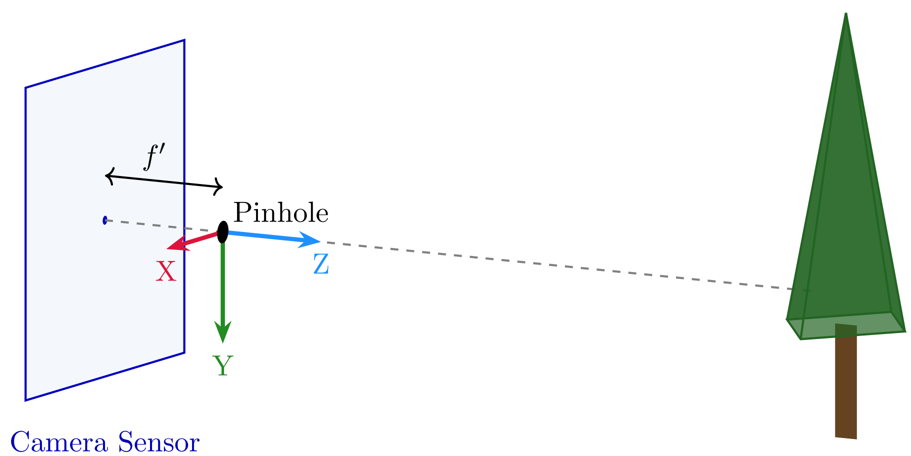
This section covers the pinhole camera model and RGB color.

1 Warmup

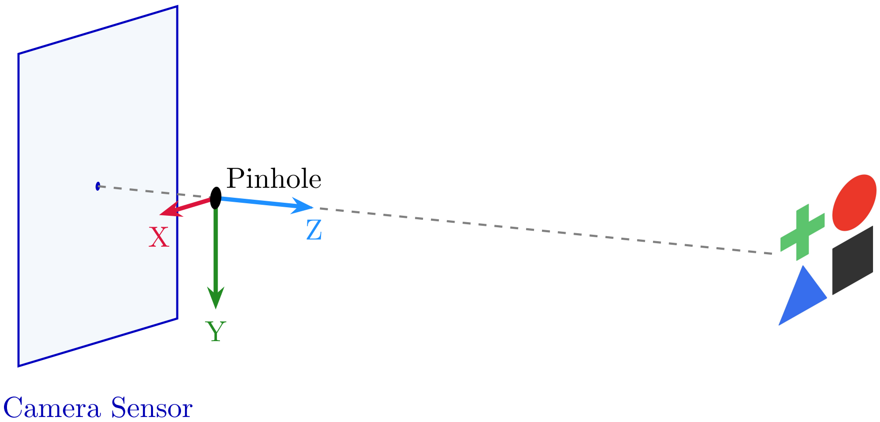
Problem 1.1: Chung Min points a pinhole camera with focal length f directly at a beautiful ponderosa pine tree. How is this object projected onto the image? Draw on the sensor below.



Problem 1.2: Unhappy with the original composition, Chung Min shortens the focal length of the camera. How does the image change? Draw on the sensor below.



Problem 1.3: Let's try on a harder object. Draw the image as it appears on the sensor below.



Problem 1.4: Assume the object has a height of H, and is located distance d away from the camera. Given focal length f, how tall is the projected object on the image plane?

$\frac{fH}{d}$

2 Dolly Zoom

In Project 0, we saw the dolly zoom effect, where multiple camera parameters are adjusted to keep a subject the same size in the image.

Problem 2.1: ***Project 0 recap.*** Between Problem 1.1 and 1.2, Chung Min decided to shorten the camera's focal length. In words, what else needs to be done to achieve a dolly zoom effect?

Shortening focal length means zoom out. To keep subject size the same, distance needs to subject needs to be decreased.

Problem 2.2: ***How to zoom?*** A camera has initial focal length f, and is placed a distance d from a subject. If the camera is moved to distance d', what focal length f' would maintain the subject's size in the image?

This can be solved by drawing similar triangles, or algebraically. One option is to start from the answer to 1.4:

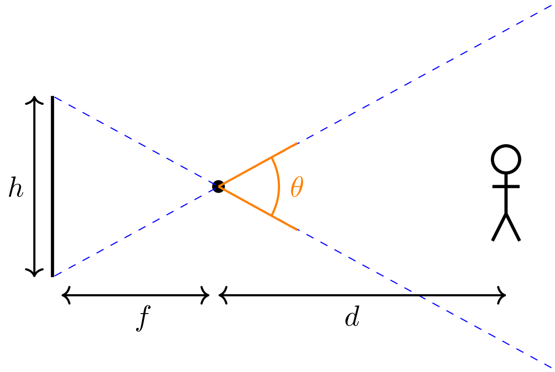
$$\frac{fH}{d} = \frac{f'H}{d'}$$

Then, to solve for f': $f' = f\frac{d'}{d}$

Problem 2.3: ***Zoom vs crop.*** Can you achieve the dolly zoom effect without changing the physical focal length of the camera system? Assume $d' > d$.

Yes, you can just crop the image.

Problem 2.4: ***Angles.*** Real-world computer vision systems often have to grapple with multiple conventions for camera geometry. One that we saw in lecture was field-of-view, which can be expressed in radians as θ:



Given initial field-of-view θ, initial subject distance d, updated subject distance d', how can we compute updated FOV θ' to achieve the dolly zoom effect?

Hint: you can start from the answer to 2.2, but your final answer should not depend on f or h.

Tangent relationship (you can draw a midline):

$$\tan(\theta/2) = \frac{h}{2}/f$$

Rearrange to get f and θ:

$$f = \frac{h}{2\tan(\theta/2)}$$
$$\theta = 2\arctan\left(\frac{h}{2f}\right)$$

These same relationships also hold for θ' and f'.

$$\begin{aligned}
\theta' &= 2\arctan\left(\frac{h}{2f'}\right) \\
&= 2\arctan\left(\frac{h}{2}\left(f\frac{d'}{d}\right)^{-1}\right) \\
&= 2\arctan\left(\frac{h}{2}\left(\frac{h}{2\tan(\theta/2)}\frac{d'}{d}\right)^{-1}\right) \\
&= 2\arctan\left(\tan\left(\frac{\theta}{2}\right)\frac{d}{d'}\right)
\end{aligned}$$

3 Color and RGB

A digital color image stores three numbers per pixel: (R, G, B). Each channel, viewed on its own, is just a grayscale image that measures how much of that primary is present.

Problem 3.1: ***Channel detective.*** Chung Min photographs a scene containing a red stop sign, green grass, and white clouds. If we display each channel as its own grayscale image, how bright does each object appear in the R, G, and B channels? Fill in the table with *bright* or *dark*.

	R channel	G channel	B channel
Red stop sign	bright	dark	dark
Green grass	dark	bright	dark
White clouds	bright	bright	bright

A saturated red object has a large R value and small G, B values, so it shows up bright only in the R channel (similarly for green). White means all three primaries are high, so the clouds are bright in every channel. Takeaway: a channel is not "colored" — it is a grayscale intensity map for one primary.

Problem 3.2: ***Fair and unfair averages.*** A common way to convert a color image to grayscale is the weighted sum

$$Y = 0.299\,R + 0.587\,G + 0.114\,B.$$

Why weight the channels unequally, instead of using the simple average $(R + G + B)/3$?

The human eye is not equally sensitive to all wavelengths: perceived brightness is dominated by green light, with red next and blue contributing the least. The weights approximate this luminance response, so the grayscale image preserves how bright the scene *looks* to us. A plain average would, e.g., make a pure blue region appear as bright as a pure green region of equal intensity, even though the green one looks much brighter to a human.